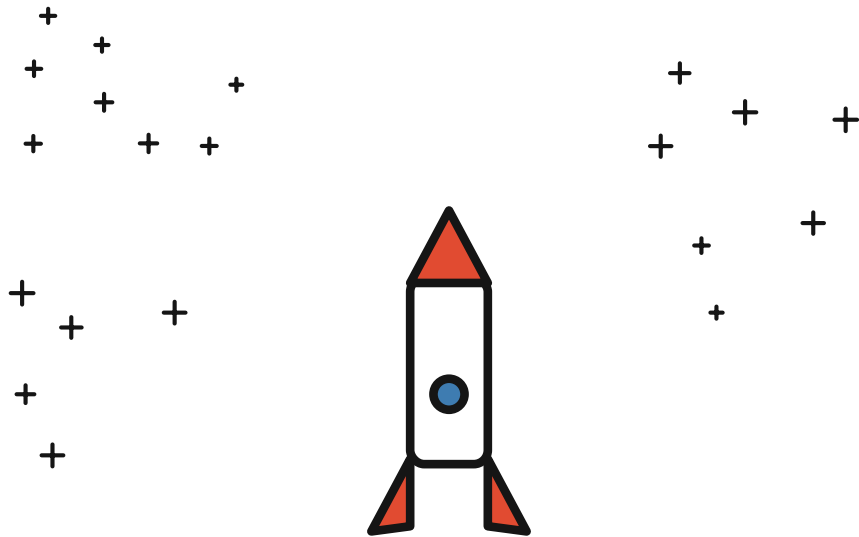


Next:

The Air Runs Out

Book Two · why you cannot fly to the Moon



Wings need air. Fires need air.
Up there, there is none of it.

LOOK UP

Book One · How far away is everything?



Print it. Read it together. Then go outside.

This is you.

It is night. Everything is quiet.



For the grown-up.

What this book is doing

It is not teaching the solar system. It is teaching one idea: that you cannot tell distance by looking, and that the sky is unimaginably deep. Everything in the rest of the series — air, thrust, orbit, staging — only matters once a child feels how far away things are.

The numbers, if he asks

The Moon averages 384,400 km away. The Sun is about 150 million km. Proxima Centauri is roughly 40 trillion km, which light itself takes 4.2 years to cross. The driving times assume 100 km/h without ever stopping.

The thumb trick is real science

Holding a finger up and blinking each eye is parallax. In 1838 Friedrich Bessel measured the same tiny shift in the star 61 Cygni as Earth moved around the Sun, and got the first true distance to another star. Before that, nobody on Earth knew.

What is coming next

Book Two asks why we cannot simply fly up there. The answer — that the air runs out, and that both wings and fires need it — is what makes a rocket a rocket.

Try this tonight.

1 Go outside and find the Moon.

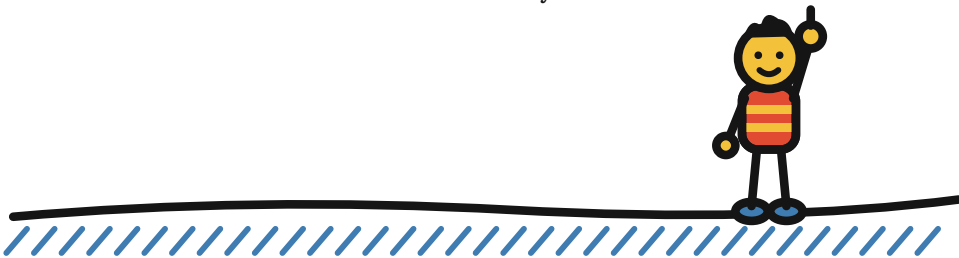
Hold your thumb up at arm's length. Cover it completely. Now you know how small it looks — and how big it really is.

2 Blink one eye, then the other.

Hold a finger up close. It jumps. Now hold it far away — it jumps less. That jump is exactly how we measured the first star.

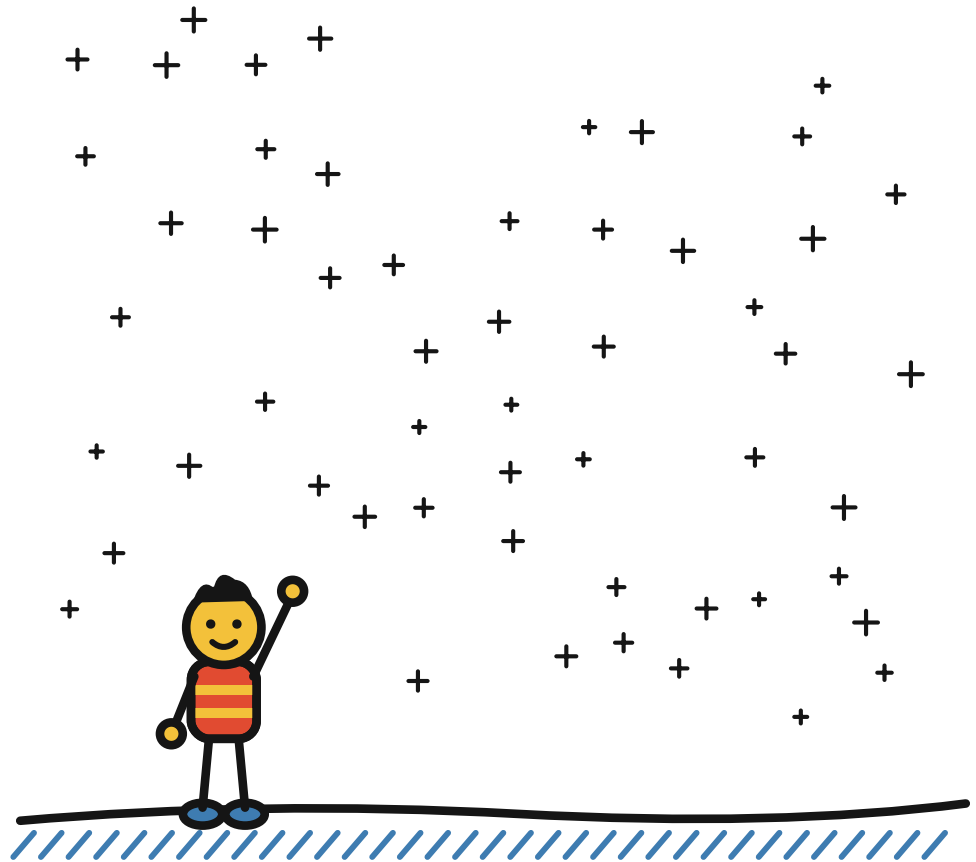
3 Pick the brightest light in the sky.

Ask him whether it is close or far. Nobody knew the answer to that for thousands of years.

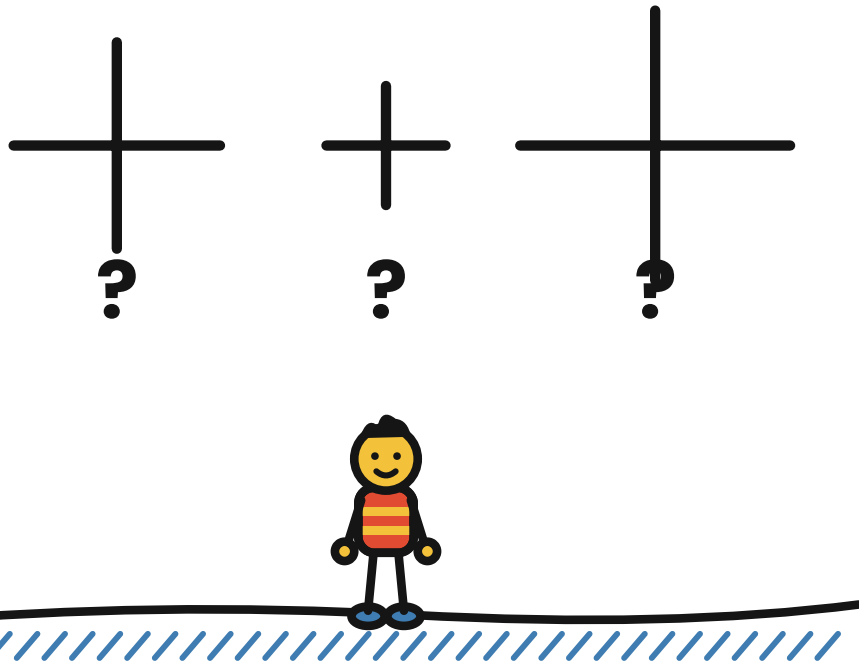


Look up.

The sky is full of little lights.



Some look big.
Some look small.

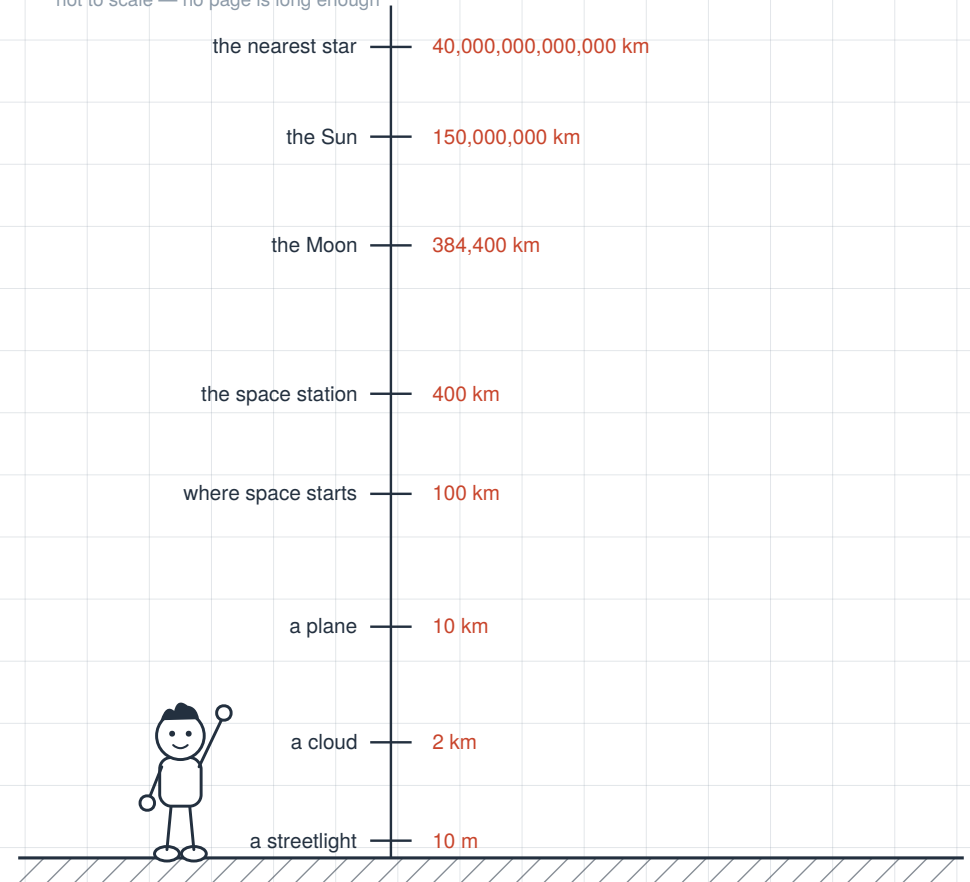


You cannot tell which one is close.

Everything, measured.

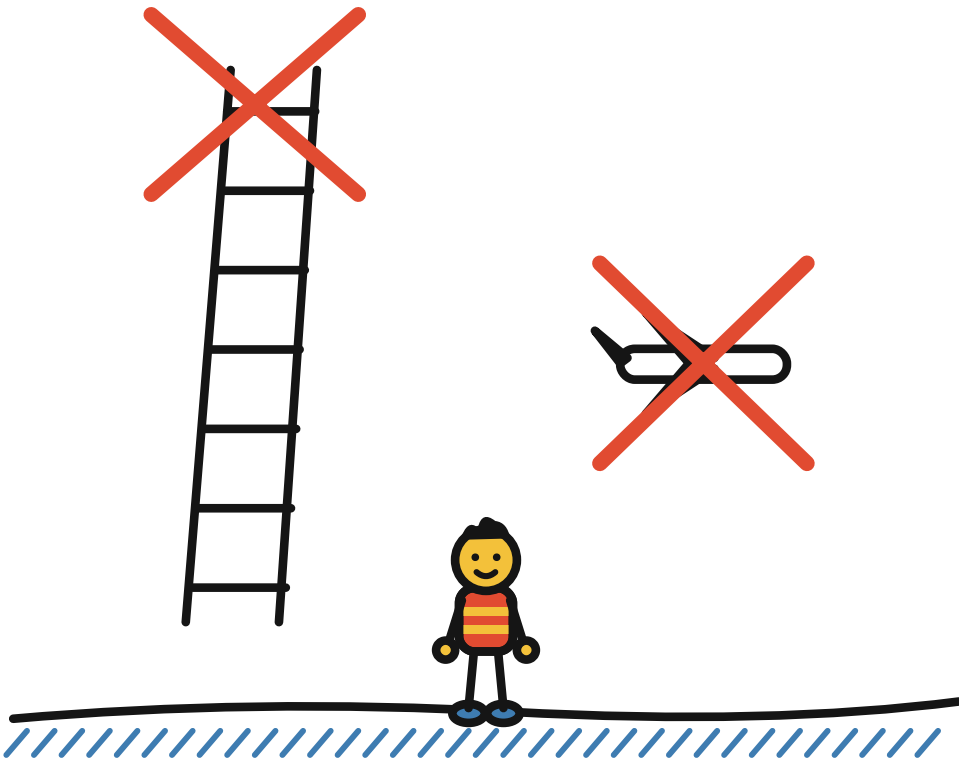
Look Up · figure 2 · the whole ladder

not to scale — no page is long enough



Each step down the page is roughly ten times the one before it.

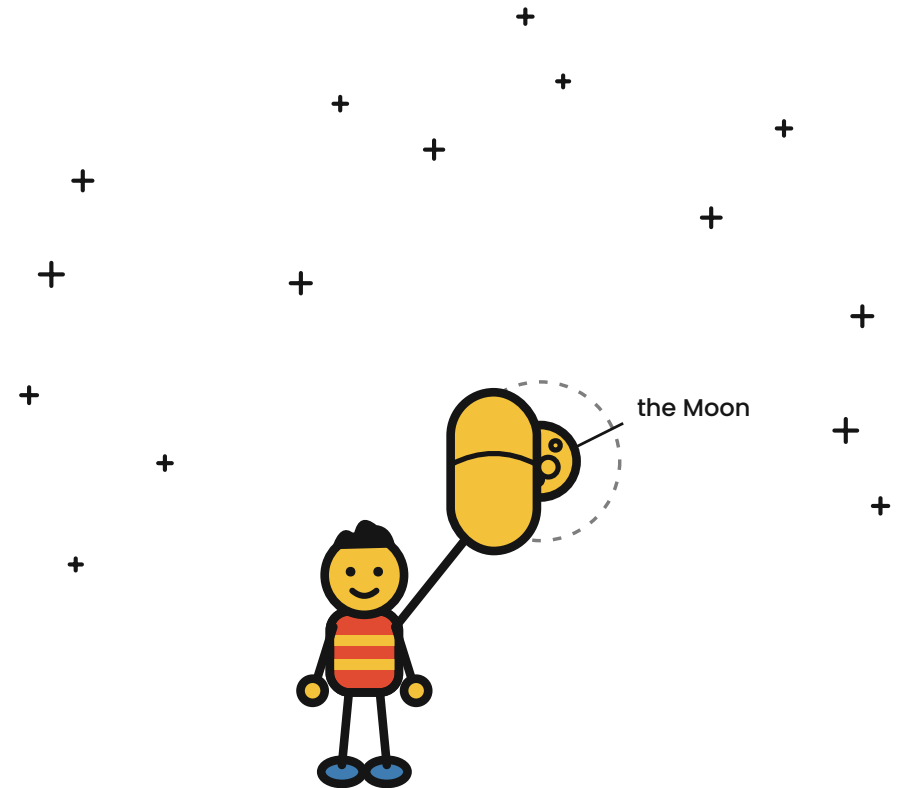
You cannot walk there.
You cannot fly there.



So people had to invent something else.

Hold up your thumb.

It covers the whole Moon.



Some things are near.



a plane · 10 km



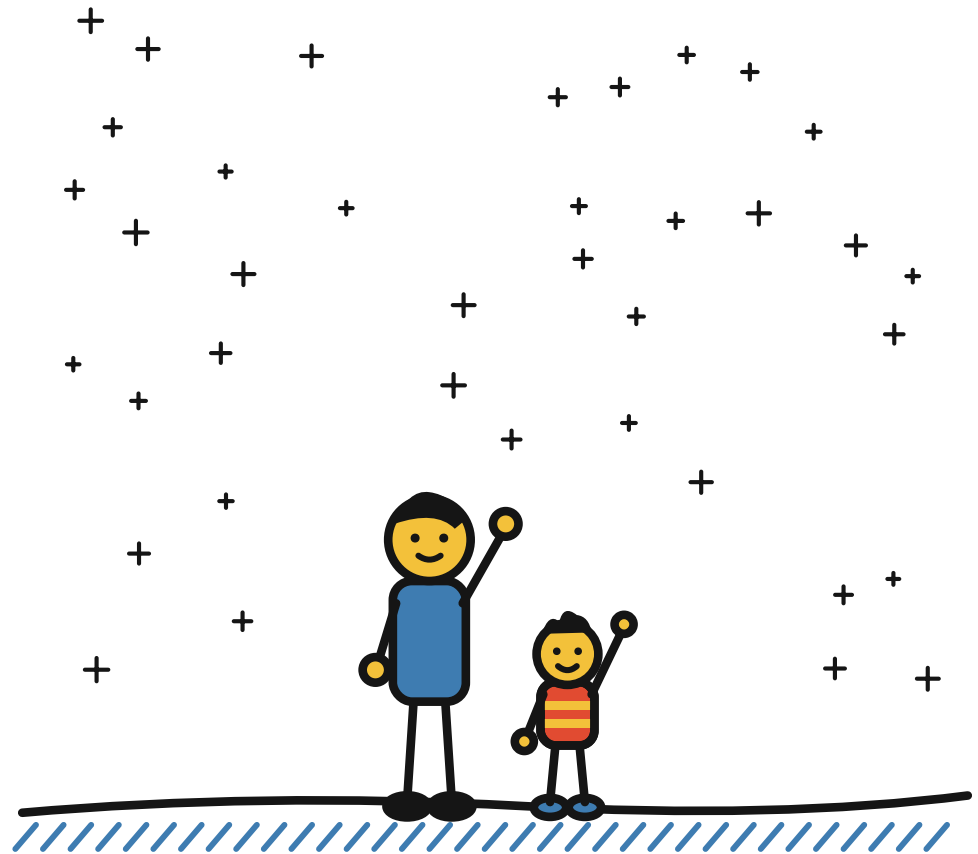
a cloud · 2 km



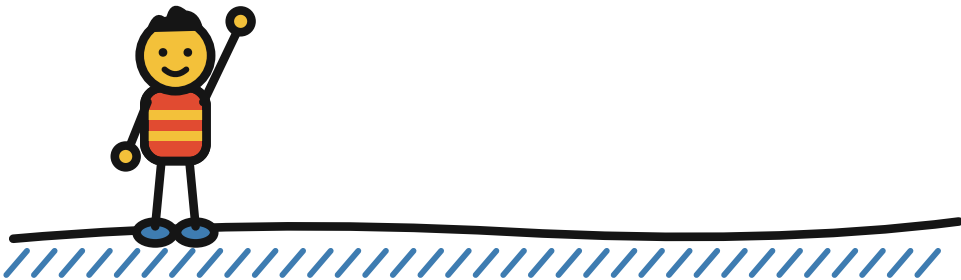
a streetlight · 10 metres

You could walk to all of them.

**The sky is not a roof.
It is a long, long way.**



And the nearest star?



Drive there and you arrive
forty-five million years later.

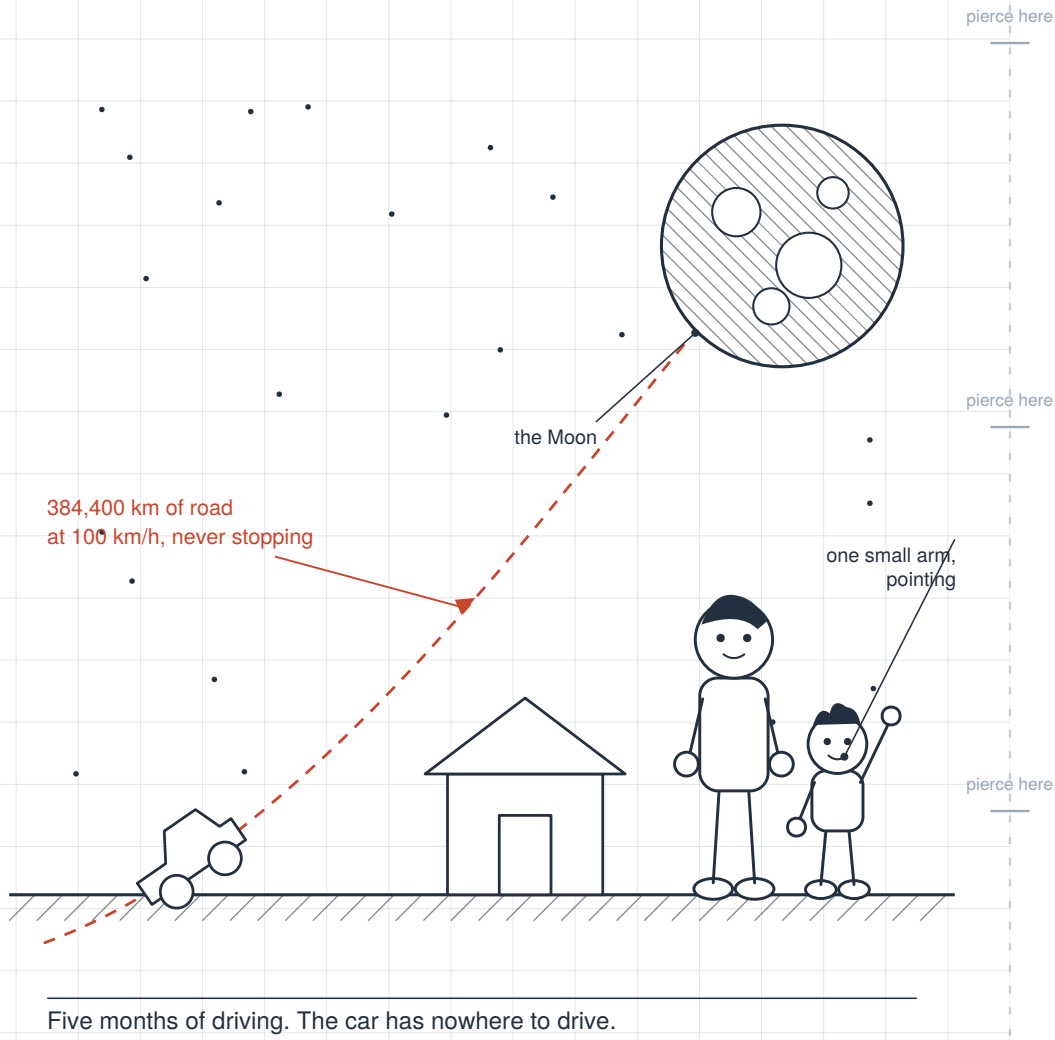
The Moon looks close. **It is not.**



Drive there without stopping: five months.

The same picture, measured.

Look Up · figure 1 · Earth to Moon



8

The Sun is farther.



9